

Information and Communication Technologies (ICT)

Information and Communication Technologies (ICTs) is a broader term for Information Technology (IT), which refers to all communication technologies, including the internet, wireless networks, cell phones, computers, software, middleware, video-conferencing, social networking, and other media applications and services enabling users to access, retrieve, store, transmit, and manipulate information in a digital form.

What are the main components of information and communication technology?

There are six main components of an ICT system. They are:

- **Data:** Facts and figures enable information and communication systems to function by providing the information they require to generate an output. Data also helps ICT users and creators build a context that gives meaning to their actions.
- **Information:** Users and systems process raw data for specific purposes, converting it into usable forms.
- **Hardware:** These are the physical components that make information and communication technology possible. Some common pieces of hardware that are a part of ICT are processors, memory cards, sound and video cards, keyboards, webcams and microphones.
- **Software:** These are the computer programs that instruct the hardware to function in predetermined ways that enable ICT. Software developers usually build them using specific programming languages and each software application has a specific role.
- **Procedures:** The people who use information and communication technology and the software applications that help them do it usually require specific procedures that outline the series of actions that each must perform to enable an ICT system to run as planned.
- **People:** The final crucial components of an ICT system are the people who use it. They use raw data to get the information they require, use hardware components like keyboards to input information into dedicated software applications and then use specific procedures

to send that information to people in different geographical locations connected through an ICT system.

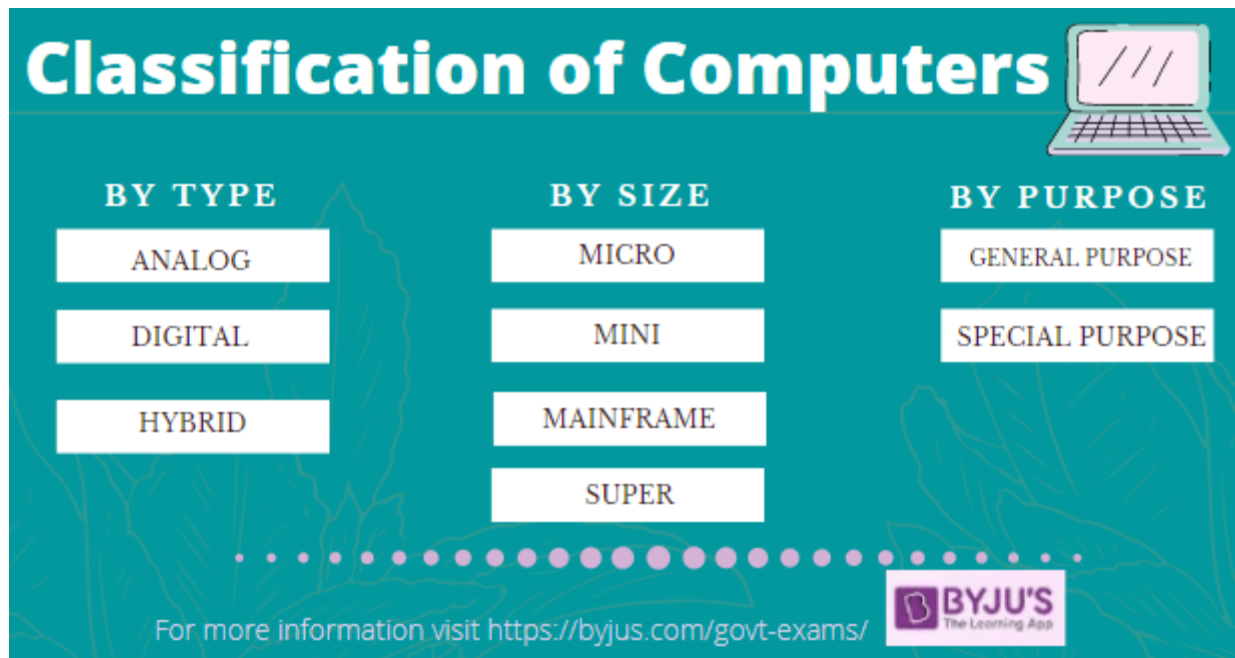
Classification of Computers

A computer is a device that transforms data into meaningful information. It processes the input according to the set of instructions provided to it by the user and gives the desired output.

There are three major categories based on which computers can be classified. These are:

1. Based on Size
2. Based on Purpose
3. Based on Types

The image given below gives a clear classification of the Types of Computers:



On the basis of data handling capabilities, the computer is of *three* types:

- Analogue Computer
- Digital Computer
- Hybrid Computer

1) Analogue Computer

Analogue computers are designed to *process analogue data*. Analogue data is continuous data that changes continuously and cannot have discrete values. We can say that analogue computers are used where we don't need exact values always such as speed, temperature, pressure and current.

Analogue computers directly accept the data from the measuring device without first converting it into numbers and codes. They measure the continuous changes in physical quantity and generally render output as a reading on a dial or scale. *Speedometer* and *mercury thermometer* are examples of analogue computers.

Advantages of using analogue computers:

- It allows real-time operations and computation at the same time and continuous representation of all data within the range of the analogue machine.
- In some applications, it allows performing calculations without taking the help of transducers for converting the inputs or outputs to digital electronic form and vice versa.
- The programmer can scale the problem for the dynamic range of the analogue computer. It provides insight into the problem and helps understand the errors and their effects.

Types of analogue computers:

- **Slide Rules:** It is one of the simplest types of *mechanical analogue computers*. It was developed to perform *basic mathematical calculations*. It is made of two rods. To perform the calculation, the hashed rod is slid to line up with the markings on another rod.
- **Differential Analysers:** It was developed to perform *differential calculations*. It performs integration using wheel-and-disc mechanisms to solve differential calculations.
- **Castle Clock:** It was invented by *Al-Jarazi*. It was able to save programming instructions. Its height was around 11 feet and it was provided with the display of time,

the zodiac, and the solar and lunar orbits. This device also could allow users to set the length of the day as per the current season.

- **Electronic Analogue Computer:** In this type of analogue computer, electrical signals flow through capacitors and resistors to simulate physical phenomena. Here, the mechanical interaction of components does not take place. The voltage of the electrical signal generates the appropriate displays.

2) Digital Computer

Digital computer is designed to perform calculations and logical operations at high speed. It accepts the raw data as input in the form of digits or binary numbers (0 and 1) and processes it with programs stored in its memory to produce the output. All modern computers like laptops, desktops including smartphones that we use at home or office are digital computers.

Advantages of digital computers:

- It allows you to store a large amount of information and to retrieve it easily whenever you need it.
- You can easily add new features to digital systems more easily.
- Different applications can be used in digital systems just by changing the program without making any changes in hardware
- The cost of hardware is less due to the advancement in the IC technology.
- It offers high speed as the data is processed digitally.
- It is highly reliable as it uses error correction codes.
- Reproducibility of results is higher as the output is not affected by noise, temperature, humidity, and other properties of its components.

3) Hybrid Computer

Hybrid computer has features of both analogue and digital computer. It is *fast like an analogue* computer and has memory and *accuracy like digital computers*. It can process both

continuous and discrete data. It accepts analogue signals and convert them into digital form before processing. So, it is widely used in specialized applications where both analogue and digital data is processed. For example, a processor is used in petrol pumps that converts the measurements of fuel flow into quantity and price. Similarly, they are used in airplanes, hospitals, and scientific applications.

Advantages of using hybrid computers:

- Its computing speed is very high due to the all-parallel configuration of the analogue subsystem.
- It produces precise and quick results that are more accurate and useful.
- It has the ability to solve and manage big equation in real-time.
- It helps in the on-line data processing.

On the basis of size, the computer can be of *five* types:

1) Supercomputer

Supercomputers are the ***biggest and fastest computers***. They are designed to process huge amount of data. A supercomputer can ***process trillions of instructions in a second***. It has thousands of interconnected processors.

Supercomputers are particularly used in ***scientific and engineering applications*** such as weather forecasting, scientific simulations and nuclear energy research. The first supercomputer was developed by ***Roger Cray in 1976***.

Characteristics or applications of supercomputers:

- It has the ability to decrypt your password to enhance protection for security reasons.
- It produces excellent results in animations.
- It is used for virtual testing of nuclear weapons and critical medical tests.

- It can study and understand climate patterns and forecast weather conditions. It can run in NOAA's system (National Oceanic and Atmospheric Administration) that can execute any type of simple and logical data.
- It helps in designing the flight simulators for pilots at the beginner level for their training.
- It helps in extracting useful information from data storage centres or cloud system. For example, in insurance companies.
- It has played a vital role in managing the online currency world such as stock market and bitcoin.
- It helps in the diagnosis of various critical diseases and in producing accurate results in brain injuries, strokes, etc.
- It helps in scientific research areas by accurately analysing data obtained from exploring the solar system, satellites, and movement of Earth.
- It also used in a smog control system where it predicts the level of fog and other pollutants in the atmosphere.

2) Mainframe computer

Mainframe computers are designed to support hundreds or thousands of users simultaneously. They can support multiple programs at the same time. It means they can execute different processes simultaneously. These features of mainframe computers make them ideal for big organizations like banking and telecom sectors, which need to manage and process high volume of data.

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Characteristics of Mainframe Computers:

- It can process huge amount of data, e.g. millions of transactions in a second in the banking sector.
- It has a very long life. It can run smoothly for up to 50 years after proper installation.
- It gives excellent performance with large scale memory management.
- It has the ability to share or distribute its workload among other processors and input/output terminals.
- There are fewer chances of error or bugs during processing in mainframe computers. If any error occurs it can fix it quickly without affecting the performance.
- It has the ability to protect the stored data and other ongoing exchange of information and data.

Applications of mainframe computers:

- In ***health care***, it enabled hospitals to maintain a record of their millions of patients in order to contact them for treatment or related to their appointment, medicine updates or disease updates.
- In the ***field of defence***, it allows the defence departments to share a large amount of sensitive information with other branches of defence.
- In the ***field of education***, it helps big universities to store, manage and retrieve data related to their courses, admissions, students, teachers, employees and affiliated schools and colleges.
- In the ***retail sector***, the retail companies that have a huge customer base and branches use mainframe computers to handle and execute information related to their inventory management, customer management, and huge transactions in a short duration.

3) Miniframe or Minicomputer

It is a ***midsize multiprocessing computer***. It consists of two or more processors and can support ***4 to 200 users at one time***. Miniframe computers are used in institutes and departments for tasks such as billing, accounting and inventory management. A minicomputer ***lies between***

the mainframe and microcomputer as it is smaller than mainframe but larger than a microcomputer.

Characteristics of miniframe or minicomputer:

- It is light weight that makes it easy to carry and fit anywhere.
- It is less expensive than mainframe computers.
- It is very fast compared to its size.
- It remains charged for a long time.
- It does not require a controlled operational environment.

Applications of minicomputers:

A minicomputer is mainly used to perform three primary functions, which are as follows:

- **Process control:** It was used for process control in manufacturing. It mainly performs two primary functions that are collecting data and feedback. If any abnormality occurs in the process, it is detected by the minicomputer and necessary adjustments are made accordingly.
- **Data management:** It is an excellent device for small organizations to collect, store and share data. Local hospitals and hotels can use it to maintain the records of their patients and customers respectively.
- **Communications Portal:** It can also play the role of a communication device in larger systems by serving as a portal between a human operator and a central processor or computer.

5) Microcomputer

Microcomputer is also known as a personal computer. It is a general-purpose computer that is designed for individual use. It has a microprocessor as a central processing unit, memory, storage area, input unit and output unit. Laptops and desktop computers are examples of microcomputers. They are suitable for personal work that may be making an assignment, watching a movie, or at office for office work.

Characteristics of a microcomputer:

- It is the smallest in size among all types of computers.
- A limited number of software can be used.
- It is designed for personal work and applications. Only one user can work at a time.
- It is less expensive and easy to use.
- It does not require the user to have special skills or training to use it.
- Generally, comes with single semiconductor chip.
- It is capable of multitasking such as printing, scanning, browsing, watching videos, etc.

Types of Computer – By Purpose

On the basis of purpose, there are just two variety of computers. Those two varieties have been discussed in detail below:

1. **General Purpose** – Based on General Purpose, there are these following functions which a device is expected to perform:
 1. Basic Input/Output functions
 2. Calculations
 3. Data Saving on a smaller scale
 4. General performing activities

These may include basic calculators, laptops, desktop computers, mobile phones, etc., which can help people with their basic necessary functions are included in the General Purpose computer type.

- **Special Purpose** – When a computer is designed specifically to perform a certain function, such type of computers is known as Special Purpose computer. These types may include:
 1. Thermometers to test temperature
 2. Generators to manage electricity

3. Devices used for analysing Climate Change
4. Large computers for IT Companies

The special-purpose computers are important for various Organizations and their applications are made in a way that makes the work easy and efficient.

Generations of Computer

Computers have evolved significantly over the years, and the history of computers is often divided into generations based on the technology used. Here are the five generations of computers:

1. **First Generation (1940s-1950s):** The first computers used vacuum tubes for processing and magnetic drums for storage. They were large, expensive, and unreliable. Some of the First Generation of Computers. ENIAC (Electronic Numerical Integrator and Computer). ENIAC is one of the first-generation computers and the first general-purpose electronic computer. UNIVAC (Universal Automatic Computer). Another one in the first generation of computers, UNIVAC was the first commercially available electronic computing device. It was a general-purpose computer designed for both scientific and commercial applications. EDSAC (Electronic Delay Storage Automatic Computer). Developed at the University of Cambridge, EDSAC aimed at overcoming the limitations of the previous computers and facilitating scientific calculations and research. As the name suggests, this computer overcame the limitation of storage and stored both the instructions and data in the same memory. IBM-701 and IBM-650.
2. **Second Generation (1950s-1960s):** The second generation of computers replaced vacuum tubes with transistors, making them smaller, faster, and more reliable. Magnetic core memory was also introduced, which was faster and more reliable than magnetic drums.
3. **Third Generation (1960s-1970s):** The third generation of computers used integrated circuits, which allowed for even smaller and faster computers. They also introduced magnetic disk storage and operating systems.

4. **Fourth Generation (1970s-1980s):** The fourth generation of computers saw the introduction of microprocessors, which made personal computers possible. They also introduced graphical user interfaces and networking.
5. **Fifth Generation (1980s-Present):** The fifth generation of computers is still ongoing, and is focused on artificial intelligence and parallel processing. This generation also saw the development of mobile computing and the internet.

Each generation of computers has built upon the advancements of the previous generation, leading to the computers we use today.

Who is the Father of Computer?

Father of Computer: Charles Babbage is the Father of Computers. Charles Babbage (1791-1871) was an extraordinarily talented scientist, mathematician, economist and engineer. Babbage, who was born in London in 1791, was a great mathematical genius. He was a natural inventor and invented all sorts of new products. He has been known as the inventor of the Analytical Engine. The Analytical Engine contained an ALU (arithmetic logic unit), basic flow control, and integrated memory; hailed as the first general-purpose computer concept.

History and Evolution of ICT

The concept of ICT can be traced back to the 19th century with the development of the telegraph and the telephone. These two inventions revolutionized the way people worked and interacted socially by making it possible for one person to communicate with another person in a different location in real or near-real time.

In the middle of the 20th century, radio and television introduced the concept of mass communication, and by the end of the century, the internet had become a mainstream tool for communication, commerce and entertainment. Its use continued to grow rapidly in the following years, revolutionizing the way people work and digitally transforming the way people communicate.

Year	Technological Advancement
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1837	Morse code and telegraph
1876	Telephone
1895	Wireless telegraphy
1927	Television
1947	Transistor
1951	UNIVAC I (First commercial computer)
1969	ARPANET (Foundation of the Internet)
1971	First email
1990	World Wide Web
1991	GSM (Global System for Mobile communication)
1994	Netscape Navigator (Widely-used web browser)
1998	Google
2001	Commercial 3G networks
2004	Facebook
2007	iPhone
2008	Bitcoin (First blockchain-based cryptocurrency)
2010	4G networks
2016	AlphaGo (AI system defeating world champion Go player)
2020	5G networks

2023 Zoom (350 million daily meeting attendees)

Today, broadband internet and smartphones have made it easier than ever for people to create, access, share and consume information and in the future, ICT promises even more exciting developments. 6G networks will provide faster and more reliable connectivity, generative AI systems like ChatGPT will change the way people create and acquire information and emerging technologies like quantum computing and Web3 will inspire a wide variety of new ICT tools and applications.

ICT History

Early History	Technological Advancements	Future Trends
 Telegraph	 Mobile Communication	 5G
 Telephone	 Broadband Internet	 6G
 Radio	 Unified Communication	 Generative AI
 Television	 Teleconference Technologies	 Quantum Computing
 Personal Computer	 Social Media	 Metaverse
 World Wide Web	 Video Collaboration Tools	 Web3

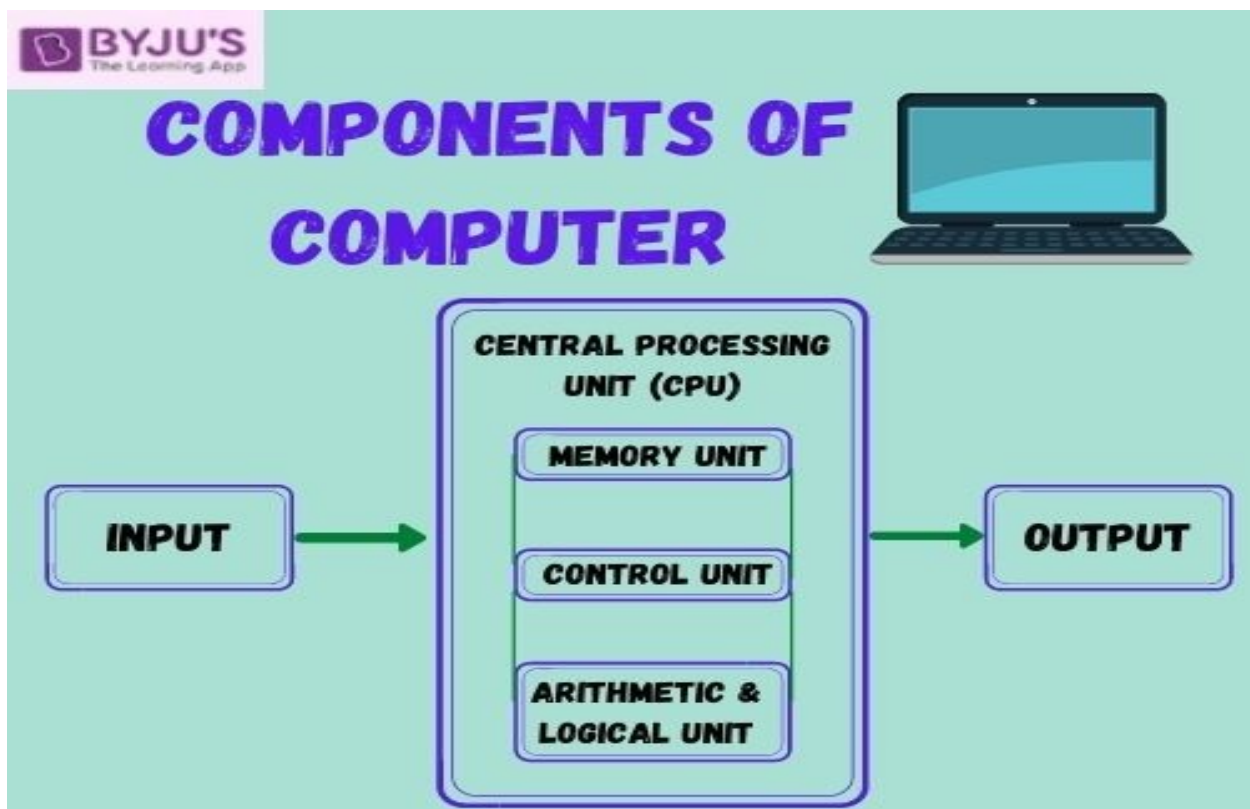
Basic Components of Computer

A computer device is made up of various elements which help in its effective functioning and processing. There are five basic components of the computer which help in making this processing of data easier and convenient.

By definition, components of a computer system are the primary elements which make the functioning of an electronic device smooth and faster. There are five basic components which include:

1. Input Unit
2. Output Unit
3. Memory Unit
4. Control Unit
5. Arithmetical and Logical Unit

5 Components of a Computer System



Given below are the 5 components of a computer along with their purpose and functions.

- **Input Unit**

A computer will only respond when a command is given to the device. These commands can be given using the input unit or the input devices.

For example: Using a keyboard we can type things on a Notepad and the computer processes the entered data and then displays the output of the same on the screen.

The data entered can be in the form of numbers, alphabet, images, etc. We enter the information using an input device, the processing units convert it into computer understandable languages and then the final output is received by a human-understandable language.

- **Output Unit**

When we command a computer to perform a task, it reverts for the action performed and gives us a result. This result is called output. There are various output devices connected to the computer. The most basic of which is a monitor. Whatever we write using a keyboard or click using a mouse, is all displayed on the monitor.

Thus, the output unit gives us the final result once the entire processing is done within the mechanism of a device.

For example: when we visit an ATM, we enter our details like language, pin, amount to be withdrawn, etc. and then the final money which the cash dispenser releases is our outcome. In this case, the cash dispenser acts as an output unit.

- **Memory Unit**

When we enter the data into the computer using an input device, the entered information immediately gets saved in the memory unit of the Central Processing Unit (CPU). Because of the presence of some existing programming, the Memory Unit transmits the data further to the other parts of the CPU.

Similarly, when the output of our command is processed by the computer, it is saved in the memory unit before giving the output to the user.

- **Control Unit**

This is the core unit which manages the entire functioning of the computer device. It is one of the most essential components of the computer system.

The Control Unit collects the data entered using the input unit, leads it on for processing and once that is done, receives the output and presents it to the user. It can be said to be the centre of all processing actions taking place inside a computer device.

Basically, the instructions taken, interpretation of entered data, issuing signals to execute the data and then finally retrieving the data is all done in the Control Unit.

- **Arithmetic & Logical Unit**

As the name suggests, all the mathematical calculations or arithmetic operations are performed in the Arithmetic and Logical Unit of the CPU.

It can also perform actions like a comparison of data and decision-making actions. The ALU comprises circuits using which addition, subtraction, multiplication, division and other numerical based calculations can be performed.

Central Processing Unit (CPU)

The Central Processing Unit is the core of any computer devices. It comprises three major components of the computer which have been discussed above:

- Memory Unit
- Control Unit
- Arithmetic and Logical Unit

All these three units are elements of CPU and together help in the efficient working and processing of data. It is also known as the “Brain of Computer” and no action can be conducted by a device without the execution and permission of the Central Processing Unit.

Software

Software is a set of programs, which is designed to perform a well-defined function. A program is a sequence of instructions written to solve a particular problem.

What is Software



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- Front End
- Back End



What is Software

Level 1

A pattern of bits given to computer hardware

```
10101101
11011010
11011010
11011010
```

1
0 > [1010 101]

Computer
Hardware

Level 2

A set of instructions given to computer

C1
Computer
program

x = 2

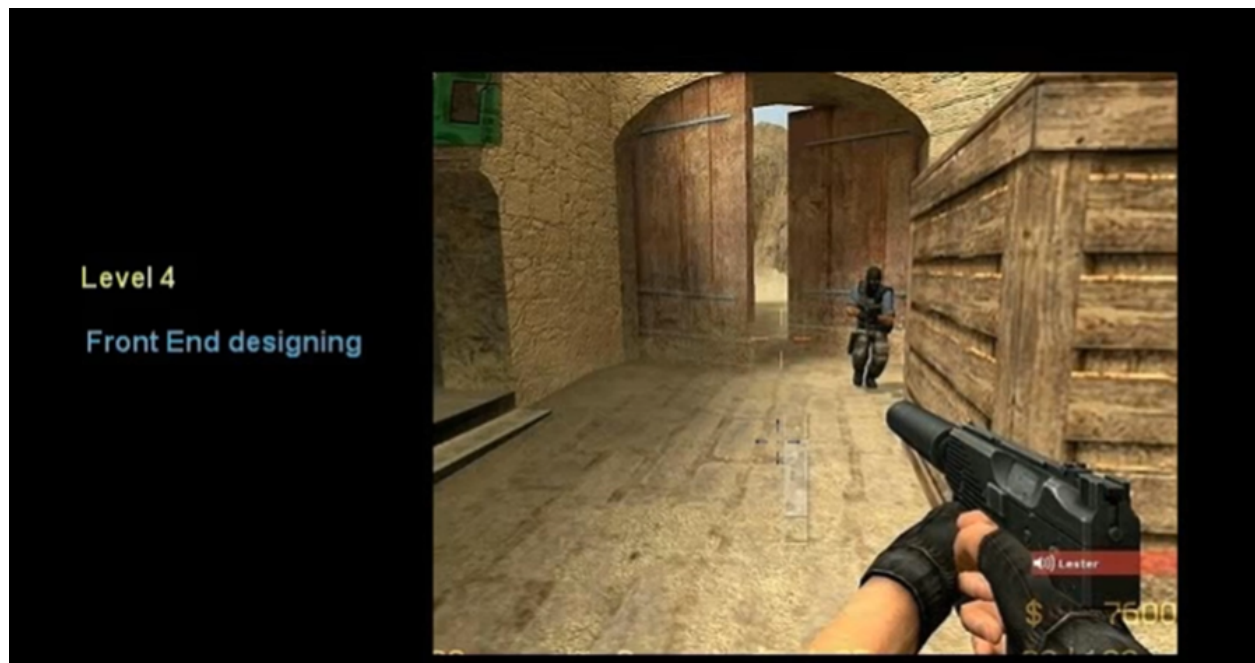
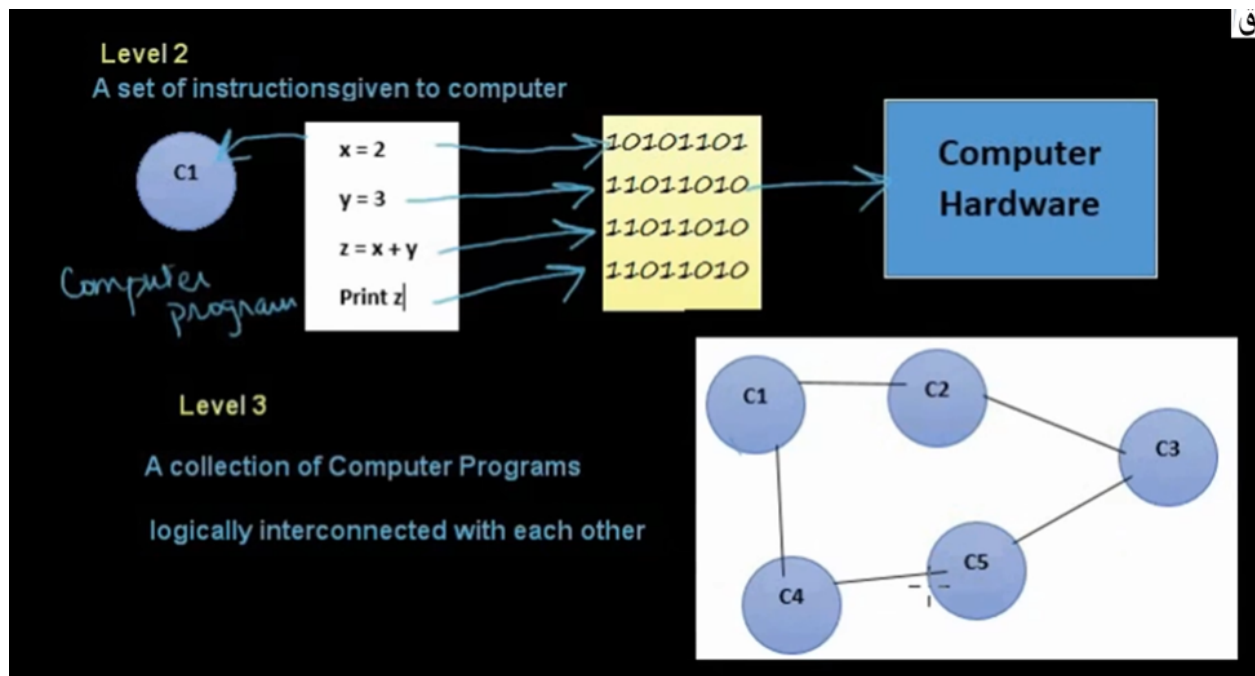
y = 3

z = x + y

Print z

```
10101101
11011010
11011010
11011010
```

Computer
Hardware



There are two types of software –

- System Software
- Application Software

System Software

The system software is a collection of programs designed to operate, control, and extend the processing capabilities of the computer itself. System software is generally prepared by the computer manufacturers. These software products comprise of programs written in low-level languages, which interact with the hardware at a very basic level. System software serves as the interface between the hardware and the end users.

Some examples of system software are Operating System, Compilers, Interpreter, Assemblers, etc.



Application Software

Application software products are designed to satisfy a particular need of a particular environment. All software applications prepared in the computer lab can come under the category of Application software.

Application software may consist of a single program, such as Microsoft's notepad for writing and editing a simple text. It may also consist of a collection of programs, often called a software package, which work together to accomplish a task, such as a spreadsheet package.

Examples of Application software are the following –

- Payroll Software
- Student Record Software
- Inventory Management Software
- Income Tax Software
- Railways Reservation Software

- Microsoft Office Suite Software
- Microsoft Word
- Microsoft Excel
- Microsoft PowerPoint



Number System in Computer

We communicate using words and characters. We are familiar with numbers, characters, and words. However, this type of data is incompatible with computers. Computers can only comprehend numbers. As a result, when we enter information, it is converted into an electronic pulse. ASCII converts the code from each pulse into a numeric format. It assigns a numerical value (number) to each number, character, and symbol. So, to understand computer language, one must be familiar with number systems. Here we explore more about the number system in computers, types of the number system in computers, and conversion of the number system in computers.

Number System in Computer

Number system in computer The most commonly used number system is the binary system, also known as the base-2 number system. The binary system uses only two digits, 0 and 1, to represent all numbers and data.

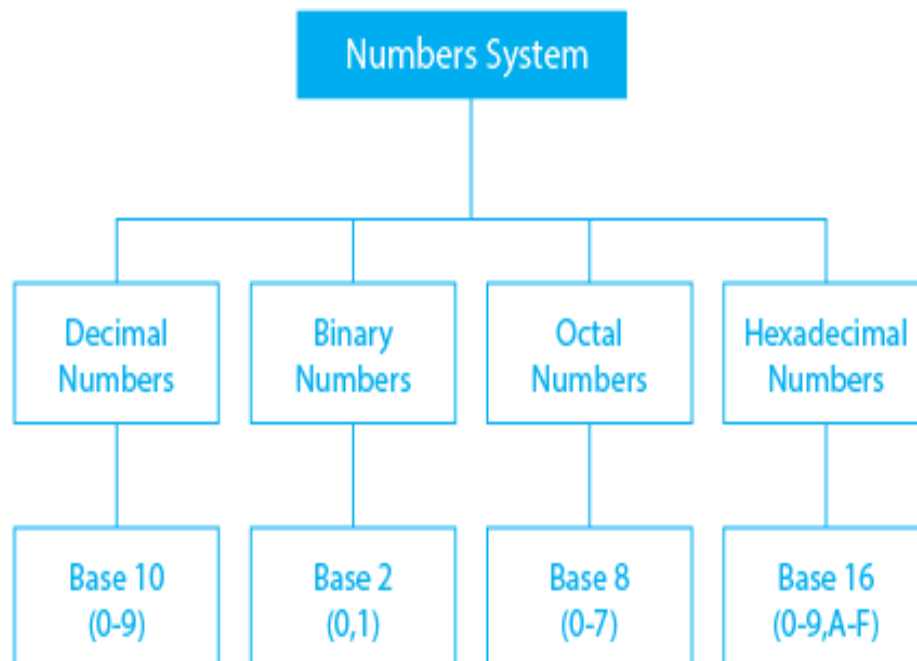
Binary numbers are used extensively in digital electronics and computing because it is easy to represent them using switches, where 0 represents "off" and 1 represents "on." This makes it possible to store and manipulate data electronically using digital circuits.

These systems are often used for representing and manipulating large binary numbers or for expressing memory addresses in a compact and easy-to-read format. Computer programs and programming languages use number system concepts to perform arithmetic operations, store data, and communicate with hardware devices.

Types of the Number System in Computer

There are mainly four types of the number system in computer

Types of Number System



Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Conversion of the Number System in the Computer

Binary Conversion:

To convert a decimal number to binary, we divide the decimal number by 2 repeatedly and write the remainder in reverse order.

$$158 / 2 = 79 \text{ remainder } 0$$

$$79 / 2 = 39 \text{ remainder } 1$$

$$39 / 2 = 19 \text{ remainder } 1$$

$$19 / 2 = 9 \text{ remainder } 1$$

$$9 / 2 = 4 \text{ remainder } 1$$

$$4 / 2 = 2 \text{ remainder } 0$$

$$2 / 2 = 1 \text{ remainder } 0$$

$$1 / 2 = 0 \text{ remainder } 1$$

Therefore, the binary equivalent of 158 is 10011110.

Convert the fractional part of decimal to binary equivalent

$$(4.47)_{10} = (?)_2$$

Step 1: Conversion of 4 to binary

1. $4/2$: Remainder = 0 : Quotient = 2

2. $2/2$: Remainder = 0 : Quotient = 1

3. $1/2$: Remainder = 1 : Quotient = 0

So equivalent binary of integral part of decimal is 100.

Step 2: Conversion of .47 to binary

1. $0.47 * 2 = 0.94$, Integral part: 0

2. $0.94 * 2 = 1.88$, Integral part: 1

3. $0.88 * 2 = 1.76$, Integral part: 1

So equivalent binary of fractional part of decimal is .011

Step 3: Combined the result of step 1 and 2.

Final answer can be written as:

$$100 + .011 = 100.011$$

Octal Conversion:

To convert a decimal number to an octal, we divide the decimal number by 8 repeatedly and write the remainder in reverse order.

$$158 / 8 = 19 \text{ remainder } 6$$

$$19 / 8 = 2 \text{ remainder } 3$$

$$2 / 8 = 0 \text{ remainder } 2$$

Therefore, the octal equivalent of 158 is 236.

Hexadecimal Conversion:

To convert a decimal number to hexadecimal, we divide the decimal number by 16 repeatedly and write the remainder in reverse order. For remainders greater than 9, we use letters A-F.

$$158 / 16 = 9 \text{ remainder } 14 \text{ (E)}$$

$$9 / 16 = 0 \text{ remainder } 9$$

Therefore, the hexadecimal equivalent of 158 is 9E.

Other Base System to Decimal System

Steps

- **Step 1** – Determine the column (positional) value of each digit (this depends on the position of the digit and the base of the number system).
- **Step 2** – Multiply the obtained column values (in Step 1) by the digits in the corresponding columns.
- **Step 3** – Sum the products calculated in Step 2. The total is the equivalent value in decimal.

Example

Binary Number – 11101_2

Calculating Decimal Equivalent –

Step	Binary Number	Decimal Number
Step 1	11101_2	$((1 \times 2^4) + (1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0))_{10}$
Step 2	11101_2	$(16 + 8 + 4 + 0 + 1)_{10}$
Step 3	11101_2	29_{10}

Binary Number – 11101_2 = Decimal Number – 29_{10}

$(0.11)_2 = (?)_{10}$

$0.11 = (1 \times 2^{-1}) + (1 \times 2^{-2}) = 0.5 + 0.25 = 0.75_{10}$

Other Base System to Non-Decimal System

Steps

- **Step 1** – Convert the original number to a decimal number (base 10).
- **Step 2** – Convert the decimal number so obtained to the new base number.

Example

Octal Number – 25_8

Calculating Binary Equivalent –

Step 1 – Convert to Decimal

Step	Octal Number	Decimal Number
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Binary addition

Binary Addition Table

The table of adding two binary numbers 0 and 1 is given below:

x	y	x+y
0	0	0
0	1	1
1	0	1
1	1	0 (where 1 is carried over)

You can see from the above table, x and y are the two binary numbers. So when we give the input for x = 0 and y = 0, then the output is equal to 0. When x = 0 or 1 and y = 1 or 0, then x+y = 1. But when both x and y are equal to 1, then their addition equals to 0, but the carryover number will equal to 1, which means basically $1 + 1 = 10$ in binary addition, where 1 is carry forwarded to the next digit.

Examples of Binary Addition

A few examples of binary additions are as follows:

Example 1: $10001 + 11101$

Solution:

$$\begin{array}{r} 1 \\ 1\ 0\ 0\ 0\ 1 \\ (+) 1\ 1\ 1\ 0\ 1 \\ \hline 1\ 0\ 1\ 1\ 1\ 0 \end{array}$$

Example 2: $10111 + 110001$

Binary Subtraction

Binary Subtraction Table

The subtraction of binary numbers is given by:

Binary Number	Subtraction Value
$0 - 0$	0
$1 - 0$	1
$0 - 1$	1 (Borrow 1 from the next high order digit)
$1 - 1$	0

Note:

The addition of two binary numbers 1 and 1 is 10, where we consider 0 and carry forward 1 to the next high order. But in the case of subtraction of 1 and 1, the answer is equal to 0, and nothing is carried forward.

In the case of decimal subtraction, when 1 is subtracted from 0, then we borrow 1 from the next preceding number and make it 10, and after subtraction, it results in 9, i.e. $10 - 1 = 9$. But for binary subtraction, it results in 1 only.

Binary Subtraction Rules

Rules and tricks: Binary subtraction is much easier than the decimal subtraction when you remember the following rules:

- $0 - 0 = 0$
- $0 - 1 = 1$ (with a borrow of 1)
- $1 - 0 = 1$
- $1 - 1 = 0$

Example 1:

Binary Subtraction

$$\begin{array}{r}
 111101 \\
 -100101 \\
 \hline
 11000
 \end{array}$$

$$\begin{array}{r}
 11001010 \\
 -10011011 \\
 \hline
 101111
 \end{array}$$

Binary Multiplication

Binary Multiplication Rules

Binary multiplication, like other [binary operations](#), is much easier, unlike the decimal multiplication when you remember the following multiplication rules. The rules of binary multiplication are:

- $0 \times 0 = 0$
- $0 \times 1 = 0$
- $1 \times 0 = 0$
- $1 \times 1 = 1$ [No borrow or carry method is applicable here]

Example 1:

$ \begin{array}{r} 101 \\ \times 89 \\ \hline 909 \\ 808 \\ \hline 8989 \end{array} $	=	$ \begin{array}{r} 01100101 \\ \times 01011001 \\ \hline 01100101 \\ 01100101 \\ 01100101 \\ \hline 0010001100011101 \end{array} $
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Binary Division

Binary Division Rules

The binary division is much easier than the decimal division when you remember the following division rules. The main rules of the binary division include:

- $1 \div 1 = 1$
- $1 \div 0 = \text{Meaningless}$
- $0 \div 1 = 0$
- $0 \div 0 = \text{Meaningless}$

Binary Division Examples

Example 1.

Question: Solve $01111100 \div 0010$

Solution:

Given

$01111100 \div 0010$

Here the dividend is 01111100, and the divisor is 0010

Remove the zero's in the **Most Significant Bit** in both the dividend and divisor, that doesn't change the value of the number.

So the dividend becomes 1111100, and the divisor becomes 10.

Now, use the long division method.

Example 2: Solve using the long division method: $01111100 \div 0010$

$$\begin{array}{r}
 10 \overline{) 1111100} \quad (111110 \\
 \underline{(-) 10} \\
 11 \\
 \underline{(-) 10} \\
 11 \\
 \underline{(-) 10} \\
 11 \\
 \underline{(-) 10} \\
 10 \\
 \underline{(-) 10} \\
 00 \\
 \underline{ 00} \\
 00
 \end{array}$$

- **Step 1:** First, look at the first two numbers in the dividend and compare with the divisor. Add the number 1 in the quotient place. Then subtract the value, you get 1 as remainder.
- **Step 2:** Then bring down the next number from the dividend portion and do the step 1 process again
- **Step 3:** Repeat the process until the remainder becomes zero by comparing the dividend and the divisor value.
- **Step 4:** Now, in this case, after you get the remainder value as 0, you have zero left in the dividend portion, so bring that zero to the quotient portion.

Therefore, the resultant value is quotient value which is equal to 111110

So, $01111100 \div 0010 = 111110$

Example 2: Solve using the long division method: $101101 \div 101$

Solution:

$$\begin{array}{r}
 101 \overline{) 101101} \quad (1001 \\
 \underline{(-) 101} \\
 101 \\
 \underline{(-) 101} \\
 0
 \end{array}$$

So, when you bring down the fourth bit of the dividend, it does not match with the divisor. In order to bring down the 5th and 6th bit of the dividend, add two zeros in the quotient value.